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Abstract The report is about an exercise proposed by the edX translation UBCx Phot1xSilicon Photonics Design, Fabrication and Data Analysis.

1 Introduction

Silicon Photonic is a way to miniaturize free space optical system into a silicon chip. For the current exercise, the author will present the Mach-Zehnder interferometer and Michelson interferometer layout made on luceda ipkiss software. Since there is a dedicated pdk for Siepic in luceda, it does ensure that the dimensions of the waveguide are widthxheigh 500x250 nm.

2 Theory

2.1 Modeling

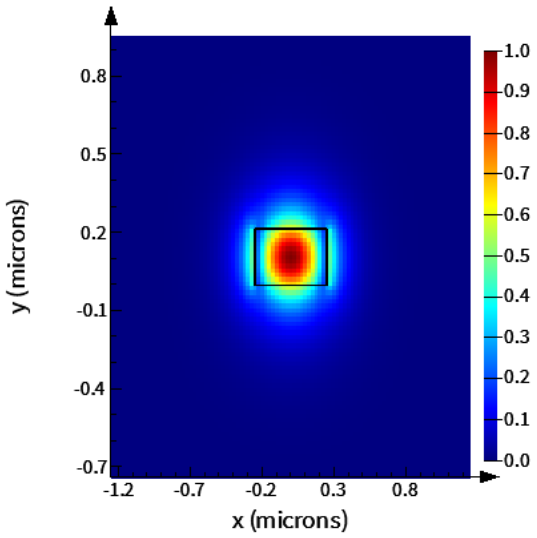

Fig. 1

TE mode in Si waveguide

2.1.1 Mach-Zehnder Interferometer (MZI)

unbalanced interferometer with identical wg

$$I_0 = \frac{I_i}{2}[1 + \cos(\beta \Delta L)] \tag{1}$$

In an MZI, light enters one port, splits, travels through the two arms once, and recombines at the output. Physical length difference: $\Delta L = L_1 - L_2$

Total path difference: Since light only goes through the arms one time,$\Delta_{total} = \Delta L = L_1 - L_2$.

FSR Formula:

$$FSR_{MZI}(\lambda) = \frac{\lambda^2}{n_g \Delta L} \tag{2}$$

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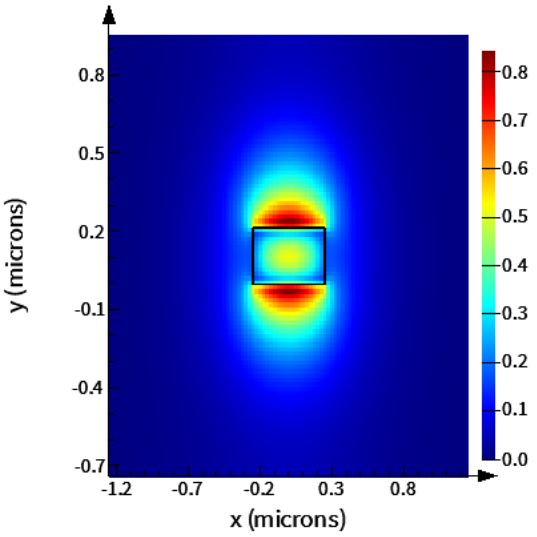

Fig. 2

TM mode in Si waveguide

2.1.2 Michelson Interferometer

In a Michelson interferometer, light splits, travels down the arms, hits mirrors, and returns back through the same arms to recombine.
Physical length difference: $\Delta L = L_1 - L_2$.

Total path difference: Because light travels the arms twice (there and back), the effective difference is doubled: $\Delta_{total} = \Delta L = L_1 - L_2$.

FSR Formula:

$$FSR_{Michelson}(\lambda) = \frac{\lambda^2}{n_g 2\Delta L} \tag{3}$$

2.2 Simulations

2.2.1 MZI with splitters

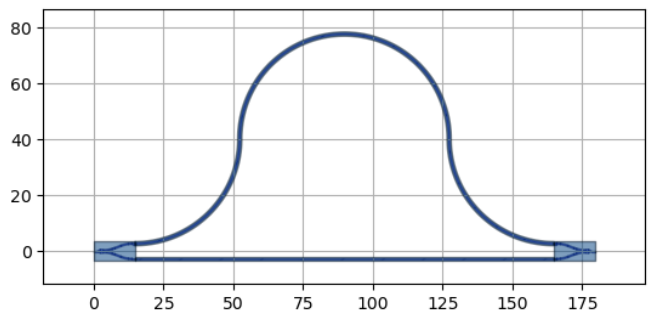

Fig. 3

MZI with splitters

2.2.2 MZI with directional couplers

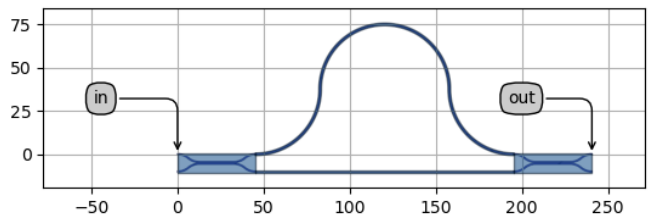

Fig. 4

MZI with directional couplers

2.2.3 Michelson Interferometer

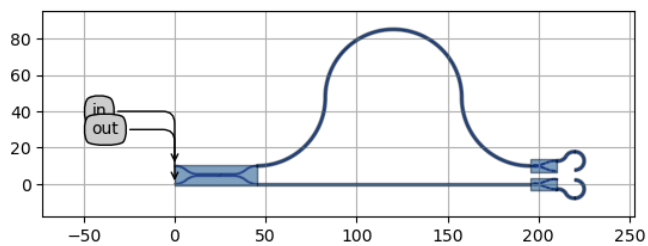

Fig. 5

Michelson Interferometer

3 Fabrication

4 Experimental Data

5 Analysis

6 Conclusion

7 Acknowledgements

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